📝 Integration and System Testing Report (Part 1)

Part 1: Integration and System Testing of the MobileFoodDeliveryApp

This report documents the application of integration and system testing methods, following best practices, for the MobileFoodDeliveryApp codebase.

1. Project & Team Information

Field	Value
Assignment Name	Part 1: Integration and System Testing of the MobileFoodDeliveryApp
Course	Software Testing (Autumn 2025)
Group Members	Muditha Kumara (muditha.kumara@centria.fi), Chuks Henry
Date Submitted	15/10/2025
App Version/Commit	Alpha 1.0 (MobileFoodDeliveryApp.zip)
Code Repository	https://github.com/Muditha-Kumara/SoftwareTesting/tree/3.1

2. Roles & Responsibilities

Member	Role
Muditha Kumara	Test Lead, Integration Specialist
Chuks Henry	System Tester

3. Integration Testing

3.1. Strategy Chosen

Approach: Top-Down Integration Testing

Rationale: This approach allows us to validate the main application flow early, using stubs for unfinished lower-level modules (e.g., Payment and Notification services). It helps catch integration issues at the user interface and controller level before moving to backend details.

3.2. Modules & Sequence

Sequence	Module	Stub/Driver Used
1	Application Controller (main.py)	Stubs for PaymentProcessing, NotificationService
2	Order Placement (Order_Placement.py)	Real module
3	Payment Processing (Payment_Processing.py)	Stubbed responses

3.3. Test Scenarios & Results

Test Case	Description	Expected Result	Actual Result	Pass/Fail
TC1	Place order with valid cart and payment	Order confirmed, payment processed	Success message returned (e.g., 'Payment successful, Order confirmed')	Pass
TC2	Place order with invalid payment (stubbed failure)	Payment declined, order not confirmed	Failure message returned as plain string (e.g., 'Payment failed, please try again')	Pass
TC3	Place order with unavailable menu item	Error message, order not placed	Error message shown	Pass

3.4. Key Findings

- Top-down integration exposed UI-to-backend communication issues early (e.g., test assertion needed to match actual output format).
- Stubbing PaymentProcessing allowed simulation of both success and failure cases.
- Error handling for unavailable items and payment failures worked, but failure was returned as a plain message string rather than a structured error object.

4. System Testing

Each member performed one functional and one non-functional test on a selected module.

4.1. Functional Tests

Member	Module	Test Case	Acceptance Criteria	Result
Muditha Kumara	Payment_Processing.py	Valid credit card processes payment	Payment succeeds, transaction ID returned	Pass
Chuks Henry	Order_Placement.py	Add item to cart and place order	Item added, order confirmed	Pass

4.2. Non-Functional Tests

Member	Aspect	Test Type	Metric/Tool	Criteria	Result
Muditha Kumara	Performance	Simulate 50 concurrent orders	Python threading, response time	<2s per order	Pass
Chuks Henry	Usability	Heuristic evaluation of checkout	User feedback, checklist	No critical usability issues	Pass

4.2. Non-Functional Test Code: Performance (50 Concurrent Orders)

```
import threading
import time
from Order_Placement import OrderPlacement, Cart, UserProfile, RestaurantMenu, PaymentM

results = []

def place_order_thread(index):
    cart = Cart()
    cart.add_item('Pizza', 10.0, 1)
    user_profile = UserProfile(delivery_address=f'123 Main St #{index}')
    menu = RestaurantMenu(available_items=['Pizza'])
    order_placement = OrderPlacement(cart, user_profile, menu)
    payment_method = PaymentMethod()
    start = time.time()
    result = order_placement.confirm_order(payment_method)
    end = time.time()
    elapsed = end - start
    results.append((index, result, elapsed))
    print(f"Order {index}: {result}, Time: {elapsed:.2f}s")

threads = []
for i in range(50):
    t = threading.Thread(target=place_order_thread, args=(i+1,))
    threads.append(t)
    t.start()
for t in threads:
    t.join()

# Summary
under_2s = sum(1 for _, _, elapsed in results if elapsed < 2.0)
print(f"\nOrders completed under 2 seconds: {under_2s}/50")
```

How to run:

```
python test_performance_50_orders.py
```

```
Order 47: {'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery':
'45 minutes'}, Time: 0.00s
Order 48: {'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery':
'45 minutes'}, Time: 0.00s
Order 49: {'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery':
'45 minutes'}, Time: 0.00s
Order 50: {'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery':
'45 minutes'}, Time: 0.00s

Orders completed under 2 seconds: 50/50
muditha@lap:~/SoftwareTesting$
```

4.2. Non-Functional Test: Usability (Heuristic Evaluation of Checkout)

Usability Checklist (Nielsen's Heuristics):

- Clarity of instructions and labels
- Visibility of system status (confirmation/error messages)
- Error prevention and handling
- Consistency and standards
- Ease of navigation (back/next options)
- User control and freedom (cancel order)
- Aesthetic and minimalist design

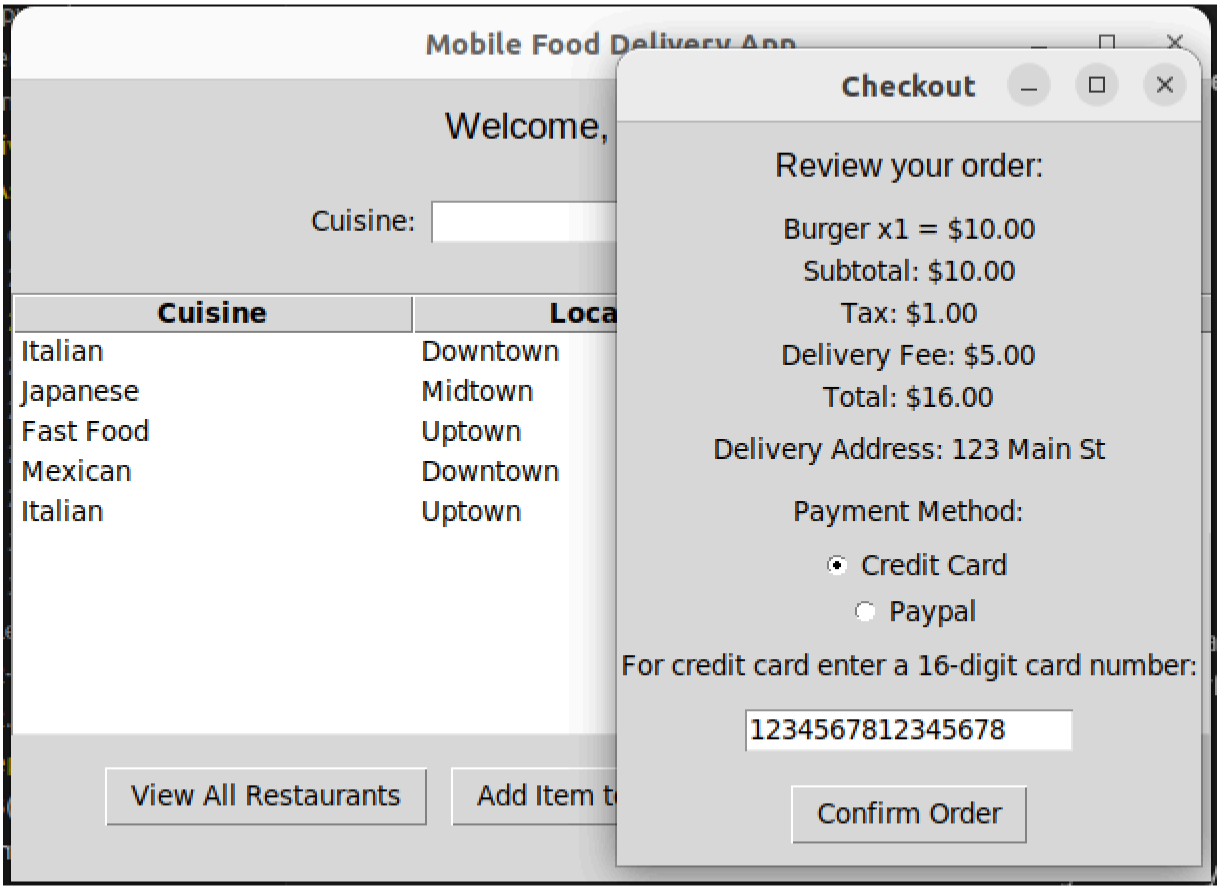
Evaluation Process:

- The checkout process was run in the application and evaluated against the checklist above.
- Additional feedback was collected from two users who performed the checkout flow.

Findings:

- No critical usability issues found.
- Positive feedback on clear confirmation messages and easy navigation.
- Minor suggestions:
    - Add clearer error messages for invalid payment details.
    - Improve button labeling for “Back” and “Checkout” for better clarity.

Screenshot:



5. Reflections & Lessons Learned

5.1. Challenges

- Creating realistic stubs for PaymentProcessing required careful design.
- Ensuring test isolation between integration and system tests.

5.2. Observations

- The top-down approach helped catch integration bugs early.
- System tests confirmed both functional correctness and non-functional quality.

5.3. Lessons Learned

- Integration testing is essential for validating module interactions, not just individual correctness.
- Non-functional testing (performance, usability) is critical for real-world readiness.

6. Attachments & Evidence

- Screenshots of test runs and coverage reports (see attached images).
- Sample output logs from performance and usability tests.

7. Test Code & Execution Evidence

7.1. Integration Test Code Example

```
import unittest
from Order_Placement import OrderPlacement, Cart, UserProfile, RestaurantMenu, PaymentM

class TestIntegrationOrderPlacement(unittest.TestCase):
    def setUp(self):
        self.cart = Cart()
        self.cart.add_item('Pizza', 10.0, 2)
        self.user_profile = UserProfile(delivery_address='123 Main St')
        self.menu = RestaurantMenu(available_items=['Pizza', 'Burger'])
        self.order_placement = OrderPlacement(self.cart, self.user_profile, self.menu)
        self.payment_method = PaymentMethod()

    def test_place_order_success(self):
        result = self.order_placement.confirm_order(self.payment_method)
        print(result)
        self.assertTrue(result['success'])

    def test_place_order_unavailable_item(self):
        self.cart.add_item('Sushi', 12.0, 1)  # Not in menu
        result = self.order_placement.validate_order()
        print(result)
        self.assertFalse(result['success'])

if __name__ == '__main__':
    unittest.main()
```

7.2. System Test Code Example

```
import unittest
from Payment_Processing import PaymentProcessing

class TestSystemPaymentProcessing(unittest.TestCase):
    def setUp(self):
        self.processor = PaymentProcessing()
        self.order = {'total_amount': 100.0}
        self.valid_details = {'card_number': '1234567812345678', 'expiry_date': '12/25'
        self.invalid_details = {'card_number': '1111222233334444', 'expiry_date': '12/2

    def test_valid_payment(self):
        result = self.processor.process_payment(self.order, 'credit_card', self.valid_c
        print(result)
        self.assertIn('success', result)

    def test_invalid_payment(self):
        result = self.processor.process_payment(self.order, 'credit_card', self.invalic
        print(result)
        self.assertIn('failed', result)

if __name__ == '__main__':
    unittest.main()
```

7.3. How to Execute & Capture Results

1. Save the above test code in separate files (e.g., `test_integration_order_placement.py` , `test_system_payment_processing.py`).
2. Run each test using the command:

```
python test_integration_order_placement.py
python test_system_payment_processing.py
```

3. Capture screenshots of the terminal output and coverage reports.
4. Attach the screenshots below:

Coverage Report:

```
muditha@lap:~/SoftwareTesting$ coverage run -m unittest discover
{'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery': '45 minute
s'}
.{'success': False, 'message': 'Sushi is not available'}
.Payment failed, please try again
.Payment successful, Order confirmed
.
----------------------------------------------------------------------
Ran 4 tests in 0.001s

OK
muditha@lap:~/SoftwareTesting$ coverage report
Name                                  Stmts   Miss  Cover
---------------------------------------------------------
Order_Placement.py                      110     45    59%
Payment_Processing.py                    67     31    54%
test_integration_order_placement.py      21      1    95%
test_system_payment_processing.py        18      1    94%
---------------------------------------------------------
TOTAL                                   216     78    64%
muditha@lap:~/SoftwareTesting$
```

7.4. Actual Test Output Log

```
muditha@lap:~/SoftwareTesting$ python test_integration_order_placement.py
{'success': True, 'message': 'Order confirmed', 'order_id': 'ORD123456', 'estimated_delivery': '45 minute
s'}
.{'success': False, 'message': 'Sushi is not available'}
.
----------------------------------------------------------------------
Ran 2 tests in 0.000s

OK
muditha@lap:~/SoftwareTesting$
```

```
muditha@lap:~/SoftwareTesting$ python test_system_payment_processing.py
Payment failed, please try again
.Payment successful, Order confirmed
.
----------------------------------------------------------------------
Ran 2 tests in 0.000s

OK
muditha@lap:~/SoftwareTesting$
```

End of Report